

Algorithm of Productivity Bot

Step ① - START

Step ② - Display menu option:

- ① Add Study Entry
- ② Add Task Done
- ③ View productivity score
- ④ Exit

Step ③ - Ask the user to enter a choice.

Step ④ - If choice = 1 (Add Study Entry):

- ① Enter Date (DD-MM-YYYY)
- ② Enter subject
- ③ Hours studied
- ④ open study.txt in append mode
- ⑤ Write the data in format:
(date, subject, hours)
- ⑥ Show message "Study log saved".
- ⑦ Go back to menu

Step ⑤ - If choice = 2 (Add task done)

- ① Ask the user to enter the task name
- ② open task.txt in append mode
- ③ Save the task in a new line
- ④ Show message "Task saved".
- ⑤ Go back to menu

Step ⑥ - If choice = 3 (View productivity score.)

- ① Initialize total-hours = 0
- ② Initialize total-task = 0
- ③ Open and read study.txt
 - ↳ For each line:
 - Split by comma
 - Add the hours to total-hours.

(iv) open and read tasks.txt

→ Count each non-empty line as one task.

(v) Calculate:

$$[\text{Score} = (\text{total_hours} \times 10) + (\text{total_tasks} \times 20)]$$

(vi) Display total hours, total tasks, and Productivity score.

(vii) Go back to menu.

Step 7 - If choice = 4 (Exit)!

(i) Display "Goodbye"

(ii) Stop the program.

Step 8 - Else (Invalid input)!

(i) Display "Invalid choice".

(ii) Return to menu.

Step 9 - ~~END~~ STOP

Flowchart

